

BRIGHT—Drift-Aware Demand Predictions for Taxi Networks

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Abstract—Massive data broadcast by GPS-equipped vehicles provide unprecedented opportunities. One of the main tasks in order to optimize our transportation networks is to build data-driven real-time decision support systems. However, the dynamic environments where the networks operate disallow the traditional assumptions required to put in practice many off-the-shelf supervised learning algorithms, such as finite training sets or stationary distributions. In this paper, we propose BRIGHT: a drift-aware supervised learning framework to predict demand quantities. BRIGHT aims to provide accurate predictions for short-term horizons through a creative ensemble of time series analysis methods that handles distinct types of concept drift. By selecting neighborhoods dynamically, BRIGHT reduces the likelihood of overfitting. By ensuring diversity among the base learners, BRIGHT ensures a high reduction of variance while keeping bias stable. Experiments were conducted using three large-scale heterogeneous real-world transportation networks in Porto (Portugal), Shanghai (China), and Stockholm (Sweden), as well as with controlled experiments using synthetic data where multiple distinct drifts were artificially induced. The obtained results illustrate the advantages of BRIGHT in relation to state-of-the-art methods for this task.

Index Terms—Time-series forecasting, concept drift, ensemble learning, global positioning system (GPS) data, mobility intelligence, taxi passenger demand, machine learning

1 INTRODUCTION

TAXI is one of the major transportation modes due to its personalized and efficient point-to-point service. Matching taxi supply with demand is one of the biggest challenges in this industry. There are two major issues: (i) the profitability requirements of the service suppliers and (ii) the *dynamic nature* of the urban mobility behavior. Drivers need to follow regulations in order to deliver high quality services to their clients while keeping/maximizing considerable profit expectations. On the other hand, passengers want to minimize the door-to-door travel times while maintaining/minimizing their costs. Recently, the increasing large-scale availability of historical taxi operational data has encouraged different exploitations of this supply/demand problem [5].

Literature traditionally addresses this problem as a recommender system, where typically each driver receives directional indications (i.e., road, stand or zone-wise) on how he should drive his vehicle to maximize profitability—either in an individual or in a collaborative fashion [8], [13], [23], [28], [31], [32]. Their main advantage is the exploitation of the two

key insights of this data type in one shot: the drivers' mobility intelligence and the passengers' demand patterns. However, existing recommender systems are lacking in tackling the dynamic nature of the network—they are based on strong assumptions such as stationarity of demand and/or urban link travel time distributions—which are quite unrealistic in practice (see, for instance, [21]). Moreover, they are typically tested in a singular case study [8], [13], [18], [20], [23], [28], [30], [31], which enforces the idea that the reported results may not be applicable to other cities.

The optimality of the drivers' decisions is heavily dependent on the urban mobility behavior—which is highly volatile and easily affected by some unpredictable *phenomena* (here denoted as incidents), regardless of their scale. Some examples of possible incidents are car accidents/breakdowns, cultural/sport events or natural hazards. Incidents can, for instance, turn optimized routes into inefficient ones, or city zones without historically high demand values into highly profitable ones at a fast rate. In Machine Learning related literature, such conceptual changes are known as *Concept Drift* [11]. Formally, a drift refers to changes in the conditional distribution of the *output*—a target variable (e.g., taxi demand quantities in an urban airport)—given the *input*—features (e.g., historical series of precipitation in that area). This phenomenon can be folded into four types [11]: it may happen (i) suddenly/abruptly (e.g., tornado/natural catastrophe), (ii) incrementally (e.g., a traffic sensor slowly wearing off and generating less accurate information), (iii) gradually (e.g., the opening of new subway line), (iv) recurrently (e.g., massive demand caused by heavy short-term rains in their season). Each type may correspond to a different change of the explanatory model in place

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(e.g., functional dependence forms shifting from linear to non-linear ones; changes in the values of the model's parameters). Consequently, distinct phenomena may require different handling mechanisms which can minimize the model's *loss* by *adapting* it to a new concept as soon as possible. Regardless of their relevance for the operational control of taxi networks, the existing works on this topic end up adopting one-shot training schemas or *blind* adaptation ones (learners that change the model without an explicit evidence that a drift occurred), i.e., by model re-training or incremental learning of the model's parameters [18], [20], [21], [30]. This issue slows the model's adaptation, increasing considerably its loss after/during drift—which is translated into a critical revenue reduction on operations—e.g., a suboptimal positioning of the fleet regarding the current spatiotemporal demand distribution.

In this paper, we propose BRIGHT: a drift-aware supervised learning framework to predict taxi demand quantities on a given area for short-term horizons in the form of point forecasts. Such predictions can be used to optimize taxi networks in a standalone fashion or integrated into a multi-criteria recommendation system (e.g., [31]) tailored for this purpose—even if the optimization problem derived from such a system is out of this paper's scope. BRIGHT operates as an ensemble of multiple time series forecasting methods which ensures two relevant properties:

- *Diversity* of the base learners that are part of the final meta-model;
- *Non-Stationarity* of the underlying concepts, *independently* of the drift type.

These characteristics are due to two *informed* adaptation strategies (i.e., *drift-aware*) to drift that are here enclosed. BRIGHT's base learners are three univariate point forecast methods with distinct memory sizes designed to handle different patterns, plus two multivariate point forecast methods where one of them has a drift-aware *neighborhood* (i.e., other demand areas) which, consequently, may evolve over time. In a latter step, a two-stage ensemble is used where the methods are first grouped (i.e., *families*) with respect to the variance of their outputs. Then, the best model from each *family* is dynamically selected by monitoring their residuals' distribution using a change detection mechanism. Finally, the model's output is combined by averaging the base models' output with respect to their recent loss.

We evaluate the generalization error of this framework on a large-scale test bed composed of three historical real-world taxi operational data—A) 450 vehicles running in the city of Porto, Portugal; B) 1.6 k vehicles from Stockholm, Sweden, C) 5.4 k taxis from Shanghai, China and a semi-synthetic one (S), where different drifts are induced artificially. The obtained results show that BRIGHT outperforms its counterparts—*especially when exposed to drift*—thus guaranteeing a high level of convergence and potential deployment in any part of the world, regardless of the urban mobility scenario in place or its traffic incident rate.

The main contributions of this paper are as follows:

- 1) A *system* to address the demand quantity prediction problem on taxi networks that leverages on combining state-of-the-art methods and procedures on learning from non-stationary distributions;

- 2) To the best of our knowledge, the *first drift-aware multi-variate time series forecasting method*—inspired from the usage of Hoeffding inequality on other adaptive learning algorithms (see, for instance, [16], [27]);
- 3) The *largest experimental setup* ever published on this problem, which uses data from three large-scale cities to evaluate our methodology according to multiple criteria, in a *total of 18.5M of trips recorded during a total period of +3 years*;
- 4) A semi-synthetic test-bed, where a dataset can be artificially generated with similar characteristics to a real one, while still accommodating the insertion of different drift types at multiple rates.

The remainder of this paper is structured as follows. Section 2 introduces our problem formulation while Section 3 formalizes the proposed methodology to approach it. The Section 4 describes the approached case studies, as well as the resulting datasets. The Section 5 shows how we tested the methodology in concrete scenarios: first, we introduce the experimental setup and the used metrics to evaluate our models; then, the obtained results are detailed and briefly discussed. Finally, the related work is briefly revised in Section 6, followed up by concluding remarks in the Section 7.

2 PROBLEM FORMULATION

The problem considered in this paper is to forecast taxi passenger demand quantities on different spatial locations across an urban city—here denoted as Regions of Interest (ROI). Such a ROI represents a discretization of space which can be generated in multiple ways, e.g.: simple stands [20] or a uniformly sized grid decomposition [6]. Let $S = \{s_1, s_2, \dots, s_K\}$ be the set of K ROI, i.e., $\mathbb{K} = \{1, \dots, K\}$, while $Y^i = \{y_1^i, y_2^i, \dots, y_t^i\}$ denote a time series of demand quantities discretized into equal-width time bins of P -minutes where $y_t^i \in \mathbb{N}$ represents the number of services demanded in ROI i during time interval $[t, t + P]$. Formally, each $Y^i, \forall i \in \mathbb{K}$ is an infinite sequence of random variables forming a random process where y_t^i is the random variable at instant t . Let $\mathbb{Y}$ represent a K -dimensional vector of the random processes $\{Y^1, Y^2, \dots, Y^K\}$. Based on the above notation, it is possible to formulate the taxi demand prediction problem as follows:

Definition 1. The taxi passenger demand prediction problem. Given: Historical realizations of random demand processes $Y^i, \forall i \in \mathbb{K}$ and $\mathbb{Y}$ denoting the following euclidean space

$$\mathbb{Y} = \begin{pmatrix} y_1^1 & y_2^1 & \cdots & y_{t-1}^1 & y_t^1 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ y_1^i & \cdots & \cdots & y_{t-1}^i & y_t^i \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ y_1^K & \cdots & \cdots & y_{t-1}^K & y_t^K \end{pmatrix} = \begin{pmatrix} \mathbf{Y}_1 \\ \vdots \\ \mathbf{Y}_{t-j} \\ \vdots \\ \mathbf{Y}_t \end{pmatrix}^T.$$

Objective: Forecast all the K realizations of $\mathbb{Y}$ at instant $t + 1$.

Typical solutions to this problem include traditional univariate and multivariate time series analysis models [4]. However, these methods depart from two assumptions that define a random process as stationary. Those can be defined as follows:

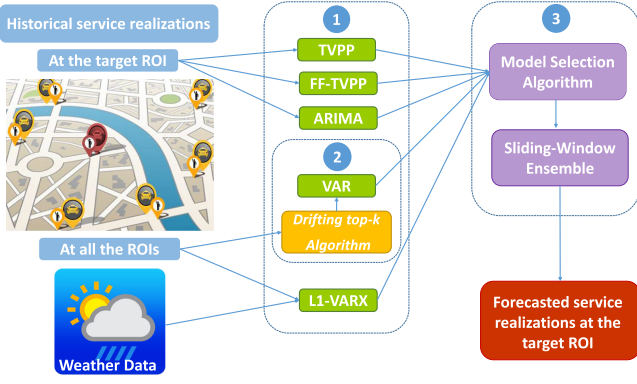


Fig. 1. Illustration of BRIGHT framework.

Condition 1. Constant Mean and Variance

$$\mu_j^i = \mu^i \wedge \sigma_j^i = \sigma^i, \forall i \in \mathbb{K}, j \in \{1..t\}.$$

Condition 2. Time-invariant Co-Variance

$$\gamma^i(t, t-j) = E[(y_t^i - \mu^i) - (y_{t-j}^i - \mu^i)] = \gamma_j^i, \forall i \in \mathbb{K}, j \in \mathbb{Z}.$$

The validity of these conditions can naturally be translated to assuming the existence of a stationary distribution $p(Y^i|X^i), \forall i \in K$ [4]. However, recent works indicate that such an assumption cannot be taken for certain (see, for instance, [21]). To handle such non-stationarity, we devise a methodology based on a hybrid drift adaptation schema.

3 METHODOLOGY

In this section, we formally introduce BRIGHT. BRIGHT operates in three different blocks: (1) first, the base learners produce demand forecasts. These are either univariate time series forecasting methods which are, in essence, individual distinct online regression models where the target variable is assumed to be a function of its historical values. Whenever dependencies with other random processes are assumed to be in place, (2) the forecasting models are denoted as multivariate. The random processes used to model the service counts value for a given ROI can be historical values from some other ROIs, denoted here as *neighborhood*, or exogenous variables (e.g., weather-related). In BRIGHT, the selection of this neighborhood can be made in a drift-aware fashion. Finally, (3) the baseline models are combined by an online meta-learner in an ensemble fashion that operates in three simple steps: first, the base learners are grouped offline using their historical outputs (e.g., using a Gaussian Mixture Model learned through EM algorithm) into at least two groups (here denoted as *families*). Second, an online model selection procedure is triggered where only one method per group (here denoted as *winners*) is exposed to the final meta-learning process. Third, each *winner's* output is combined in a weighted average where these weights are inversely proportional to their recent *loss*—similar to what is proposed by other popular *blind* adaptation ensembles [11]. The three steps of this methodology are illustrated in Fig. 1 and formally presented throughout this section.

3.1 Base Learners

The five base learners that we incorporate in our framework are as follows: (i) a Time-Varying Poisson Process (TVPP),

(ii) a Fading-Factor TVPP (FF-TVPP), (iii) an Autoregressive Integrated Moving Average (ARIMA) process, (iv) an L1-regularized Vector AutoRegressive process with exogenous variables (L1-VARX), and (v) a Drift-Aware VAR process. Methods (i-iii) are already proposed—natively or in their variants—by [20]. Here, we simply modify their training schema to improve their adaptiveness. The latter two methods (iv, v) are novel and specifically designed to this problem: (iv) puts forward a regularized version of VARX with a particular process to tune their hyperparameters online while (v) defines an incremental way of defining neighborhoods. These methods differ in memory size, input data and in the training which aims to handle different types of drift. They are formally presented below.

3.1.1 TVPP

One of the simplest probabilistic models for count observations is the Poisson distribution

$$P(n|\lambda) = \frac{e^{-\lambda} \lambda^n}{n!} : n \in \mathbb{N}, \quad (1)$$

where the parameter λ represents the constant arrival service rate in a fixed time interval. In order to model the typical seasonalities of human behavior (e.g., trips from/to home/work), we use a time-dependent service rate $\lambda(t)$, as originally proposed in [20], turning (1) into a nonhomogeneous Poisson process. Specifically, we decompose $\lambda(t)$ as follows:

$$\lambda(t) = \lambda_0 v_{d(t),h(t)}, \quad (2)$$

where $v_{d(t),h(t)}$ is the relative change for the weekday $d(t)$ (e.g.: Saturdays have lower service rates than Tuesdays) and for the period $h(t)$ on the day $d(t)$ (e.g., the peak hours); $d(t)$ represents the weekday 1 = Sunday, 2 = Monday,...; and $h(t)$ indicates the interval in which time t falls.

TVPP: Training and Drift Adaptation. Naturally, the predictions of the TVPP model are univariate, relying on time-dependent averages of past realizations, and can be computed incrementally, i.e., $\hat{Y}_{t+1}^i = \lambda^i(t)$. However, this formulation does not include any forgetting mechanism. Consequently, the adaptation happens at a slow rate—which makes the model behave similarly to an offline model, or at most, enables it to deal *blindly* with incremental drifts.

3.1.2 FF-TVPP

To address some limitations imposed by TVPP, we extend it by including fading factors that will limit such memory, thus increasing the adaptation rate

$$\hat{Y}_{t+1}^i = \sum_{j=1}^t \omega(j) \lambda^i(j) \quad (3)$$

$$\omega(j) = \begin{cases} 0 & \text{if } (1 - \alpha)^{(t-j)} < \epsilon_\omega; \\ (1 - \alpha)^{(t-j)} & \text{otherwise.} \end{cases} \quad (4)$$

where the hyperparameters $\alpha, \epsilon_\omega \in (0, 1)$ set the model's adaptation rate and its memory size, respectively.

FF-TVPP: Training and Drift Adaptation. The hyperparameter α needs to be tuned. On the other hand, ϵ_ω is only a minimum threshold to limit the memory size of the method

to samples relevant to its output. FF-TVPP is able to *timely* (but still *blindly*) adapt to gradual drifts due to its concept of recentness given by the fading factor ω . A gradual drift occurs when the underlying distribution changes by alternating between an old and a new functional form during a period of time [10]. In this application, a concrete example of such *smooth* drifts can be the access to beaches in the transition between Summer and Fall's weather—while in some days beach areas may continue to have demand peaks in late afternoon, weather oscillations—such as the first Fall' rains—will push people to leave beach on other timings and, ultimately, not even going/leaving to/from there at all.

3.1.3 ARIMA

The AutoRegressive Integrated Moving Average (ARIMA) process is a well-known methodology for both modeling and forecasting univariate time series data due to its ability to react to abrupt changes in the input signal [4]. Departing from an assumption of stationarity on every Y^i , every ARIMA process turn to be an ARMA one

$$\hat{Y}_{t+1}^i = \delta^i + \sum_{j=t-1}^{t-p} \phi_j^i Y_j^i - \sum_{j=t-1}^{t-q} \theta_j^i \epsilon_j^i + \epsilon_{t+1}^i, \quad (5)$$

where ϵ_{t+1}^i is a Gaussian noise component of the most recent process realization; ϕ^i and θ^i are vectors denoting model parameters while $p, q \in \mathbb{N}$ refer to the model's order.

ARIMA: Training and Drift Adaptation. Typically, ARMA processes are used in a *sequential* setting, where the value of each target random variable is predicted based on its past observations. When the target sample's label/value is known, it is used to re-train the model before the next prediction takes place. In order to handle drift, typical approaches include estimating the model order (p, q) automatically, using the auto-correlation and partial auto-correlation profiles as proposed in [4], and periodically (e.g., daily [18]). Similarly, the parameters ϕ^i, θ^i are also typically re-fitted offline before each prediction using a suitable optimization procedure (e.g., ordinary least squares with $L2$ -loss [18]).

Following the work in [21]—which is inspired by the delta rule typically embedded in Stochastic Gradient Descent—we propose to update the parameters ϕ^i, θ^i incrementally as

$$\phi_t^i = \phi_{t-1}^i (1 - \eta(r_{t-1}^i)) \quad (6)$$

$$\theta_t^i = \theta_{t-1}^i (1 + \eta(r_{t-1}^i)), \quad (7)$$

where $r_{t-1}^i = \hat{Y}_{t-1}^i - Y_{t-1}^i$ is the model's *loss* estimation at $t-1$ while η denotes a hyperparameter which represents a constant learning rate and, consequently, needs to be tuned. These changes allow the model to be more (*blindly*) responsive to some drift types by speeding up its adaptation rate.

The base learners that have been proposed so far are univariate. However, the demand in a given ROI can be affected by other ROIs. The drift-aware model presented next tries to address this problem.

3.1.4 VAR

The Vector Autoregressive (VAR) model is one of the most commonly applied tools to handle dependencies among multiple random processes for forecasting purposes [4].

Similarly to ARMA models, VAR also departs from a stationarity assumption and possesses an order p

$$\begin{aligned} \hat{\mathbf{Y}}_{t+1} &= \mathbf{Y}_t^T \Phi_0 + \mathbf{Y}_{t-1}^T \Phi_1 + \cdots + \mathbf{Y}_{t-p+1}^T \Phi_{p-1} + \epsilon_t \\ &= \sum_{j=1}^p \mathbf{Y}_{t-j+1}^T \Phi_{j-1} + \epsilon_t \end{aligned} \quad (8)$$

where $\Phi = (\Phi_0, \dots, \Phi_{p-1}) \in \mathbb{R}^{p \times \mathbb{K}}$ are the model's parameters while ϵ_t is a vector white noise, i.e., $E(\epsilon_t) = \mathbf{0}$.

VAR: Training and Drift Adaptation. The traditional VAR models share most of the properties, issues and training schemas abovementioned to the ARMA ones. Naturally, the parameters matrix can easily scale up to a high dimensionality. Similarly to other regression frameworks, it is expected that the model complexity increases, thus leading (with a high likelihood) to sparse solutions.

3.2 L1-Regularized VAR(X)

Weather is known to be of huge importance when devising explanatory models around transportation problems (e.g., bikesharing [9] and mass transit [19]). In order to incorporate such potential knowledge in BRIGHT, we add multiple time series of meteorological variables, denoted as $\mathbf{W}$ which can be used as explanatory features only. The resulting feature space can be defined as $\mathbf{Y}' = \mathbf{Y} + \mathbf{W}$.

Naturally, the inclusion of weather-based exogenous variables can aggravate even more the abovementioned sparsity issue. To mitigate it, we propose to apply an L1 regularization schema while learning the VAR process parameters. The intuition behind it is that the demand in each ROI only depends on a small set of its pairs plus, eventually, some exogenous factors. Together, this dependence structure forms a *neighborhood*. This neighborhood is naturally discovered by the introduction of a shrinkage scalar parameter β in the optimization problem

$$\min_{\Phi} \left\{ l(\hat{\mathbf{Y}}_{t+1}, \mathbf{Y}_{t+1}) + \frac{1}{2} \|\Phi\|^2 \right\}, \quad (9)$$

where $l(\hat{\mathbf{Y}}_{t+1}, \mathbf{Y}_{t+1})$ denotes the chosen *loss* function.

L1-VARX: Training and Drift Adaptation. This model shares the same properties and training schema as the classic VAR/ARMA models. The usage of L1-penalty aims to mitigate the overfitting likely causes since it is well suited to handle sparsity issues in the feature space. After tuning β , the model can be learned incrementally—turning it to be (*blindly*) responsive to incremental and/or gradual drifts. However, this setting may require a periodic fine tuning of β as the definition of the neighborhood (and consequent regularization) is still mainly happening in an offline setting.

3.3 Drift-Aware VAR with Top- k Selection

To realize a drift-aware regularization of a VAR model, we devise a two-stage online algorithm denoted *Drifting Top-k*. In its first stage, this procedure selects a neighborhood $\mathbb{L}$ for each target variable in every random process $Y^i, \forall i \in \mathbb{K}$ of k other processes, transforming Eq. (8) as follows:

$$\hat{Y}_{t+1}^i = \sum_{l \in \mathbb{L}} \sum_{j=1}^p Y_{t-j+1}^l \phi_{j-1}^l + \epsilon_t^l, \quad (10)$$

where $|\mathbb{L}| = k$, a user-defined hyperparameter.

This neighborhood is selected through an incremental similarity measure by the closest k processes to the target one. Hereby, we propose to use a custom measure based on the Pearson's correlation—commonly used to deal with time series data (see, for instance, [27])—denoted as *SRC*—*Scaled Root Correlation* and computed as follows:

$$\text{corr}(a, b) = \frac{\tau - \frac{AB}{n}}{\sqrt{C - \frac{A^2}{n}} \sqrt{D - \frac{B^2}{n}}} \quad (11)$$

$$\text{src}(a, b) = \sqrt{\frac{1 - \text{corr}(a, b)}{2}} \in [0, 1], \quad (12)$$

where n denotes the total number of observed samples.

Theorem 1. *In BRIGHT, the similarity measure used in the regularization schema applied within the proposed drift-aware VAR model is an incremental statistic where $\text{src}(a, b)_t = f(\text{src}(a, b)_{t-1}, a_t, b_t)$.*

Proof. *src* only depends on the correlation function, *corr*. Following Eq. (11), we conclude that $\text{corr}(a, b) = f(\tau, A, B, C, D)$. All those variables are defined as functions of a, b in a fully additive fashion, as it is demonstrated below

$$A = \sum_{j=1}^n a_j, B = \sum_{j=1}^n b_j, C = \sum_{j=1}^n a_j^2, D = \sum_{j=1}^n b_j^2 \quad (13)$$

$$\tau = \sum_{j=1}^n a_j b_j. \quad (14)$$

This completes the demonstration of Theorem 1. $\square$

The definition of the neighborhood is still constrained by minimum absolute similarity value between the target ROI and its counterparts, ξ , which is also a user-defined hyperparameter.

Naturally, the definition of this neighborhood is not static, as the dependencies among different random processes follow non-stationary concepts. The stationarity condition for this neighborhood is hereby formulated as follows.

Condition 3. *Let $Z_t^i \in \mathbb{R}^{k \times k}$ be a symmetric similarity matrix of a given neighborhood and z_t^i be a vector containing all the elements in Z_t^i where $z_{j,t}^i \geq z_{j-1,t}^i, \forall j \in \{1 \dots k^2\}$. Let v^i denote the minimum *src* coefficient of a given neighborhood at the instant of its generation t_0 . The dependence structure is said to be weakly stationary if the true mean of Δz_t^i is 0*

$$\Delta z_t^i = |z_{1,t}^i - v^i|. \quad (15)$$

The intuition behind this schema is based on an assumption that the distance between the two most dissimilar random processes within the same neighborhood sets its *boundary* under a form of a logical *diameter*. If this *boundary* diverges significantly over time, a drift is assumed to be in place. Hereby, we propose to test the validity of this condition incrementally using the well-known Hoeffding Bound [15]. The Hoeffding Bound states that after n independent observations of a real-value random variable with range R , its true mean has not diverged if the sample mean is within $\pm \epsilon_F$

$$\epsilon_F = \sqrt{\frac{R^2 \ln(1/\delta)}{2n}}, \quad (16)$$

with a probability of $1 - \delta$ (a user-defined hyperparameter). Once Condition 3 is violated, an alarm is triggered, the neighborhood composition is reorganized using Z_t^i and the reference *diameter* v^i is reset by setting $t_0 = t$ and $n = 0$.

3.4 Two-Stage Ensemble

The metalearning process of our framework operates in two stages: (1) model selection and (2) stacking through model averaging. Its training is an online process that runs at test time supported by an offline block-based initialization, i.e., iterated over a validation dataset. This procedure is described throughout this section.

3.4.1 Model Selection

One of the key ingredients for successful ensembles is *diversity*. Typically, this diversity is reflected in distinctive patterns of each model's inductive bias derived from the different assumptions that their learners pose with respect to the data input and the functional form of the dependence structure in place. In BRIGHT, the differences lie mainly in their memory size, which allows them to model different types of patterns, and in the features used to compute each individual prediction (i.e., univariate, using only the historical realizations of the same process, or multivariate, considering dependences among multiple different random processes). Surprisingly, the enforcement and evaluation of diversity on ensembles for non-stationary data is a quite unexplored topic—especially in regression problems [17].

Hereby, we propose a simple schema of Model Selection that tries to ensure such diversity through *clustering*. The base idea is to split the methods into *families* to then select one of each family—*winner*—to be part of the final model. This splitting is hereby proposed to be made using an offline clustering framework aimed to maximize the variance—e.g., a Gaussian Mixture Model solved with Expectation-Maximization algorithm. This procedure is iterated over a validation set according to the Euclidean distance among the series of outputs. The determination of the ideal number of *families* is a model selection problem that can be solved by any entropy-based metric. As the base learners' assumptions on the data are immutable (with exception to the drift-aware VAR model), the authors assume stationarity in this particular context (i.e., *families'* definition)—even if we do not discard anyhow the validity of such hypothesis.

Initially, to select each *family winner*, we basically observe the average loss of each method within the validation set and retain the model with the lowest one. However, this selection may change in run time as a consequence of observed concept drifts. For this purpose, we employ the Page-Hinkley (PH) test [22], a well-known change detection method. Its pseudo code is introduced in Algorithm 1. The PH-test has two hyperparameters: v, ϱ that need to be tuned before usage. Once a drift alarm is triggered in a given instant t_d (i.e., output of the PH-test), a new *family winner* is selected based on the loss estimation of each method over a recent fixed-size sliding window of time. Naturally, upon an alarm trigger, the error estimators of the PH-test are restarted alongside the overall algorithm.

This mechanism has two direct benefits: it reacts to drift in an informed fashion while encouraging diversity by ensuring a high variance among the predictors used in the final meta-model. Such variance derives from the expertise of each predictor, which is likely to be higher for a specific feature subspace rather than in the entire one.

Algorithm 1. Page-Hinkley Algorithm

Input: Admissible change v , Drift Threshold ϱ , Loss at example t : ϵ_t ;
Output: Drift time instant t_d

```

1 /* Initialize the error estimators */
2  $SR(0) \leftarrow 0$ ;  $m_T \leftarrow 0$ ;  $M_T \leftarrow 1$ ;
3 /* Page-Hinkley test */
4 for  $i \in \{1..\infty\}$  do
5    $SR(i) \leftarrow SR(i-1) + \epsilon_i$ ;
6    $m_T(t) \leftarrow m_T(t-1) + \epsilon_t - \frac{SR(t)}{t} - v$ ;
7    $M_T \leftarrow \min(M_T, m_T(t))$ ;
8   if  $(m_T(t) - M_T) \geq \varrho$  then
9     return  $i$ ;
10 end
11 end

```

3.4.2 Fixed-Size Sliding Window Ensemble

Adaptive ensembles are derived from the assumption that, during a drift, samples are generated from a mixture of distributions rather than a singular one [11]. Naturally, this can be seen as a weighted combination of distributions characterizing the target concept. If each model approximates a singular distribution, the true value of the target variable will be given by a weighted combination of all models' outputs.

The model that we propose hereby is denoted as Fixed-Size Sliding Window Ensemble. It is similar to other methods typically employed for this problem (e.g., [20]). Let M be a set of base learners of interest (in our case, the *family winners*) whose output of model j for ROI i in time instant t is given by $m_{j,t}^i$. The final prediction is obtained by

$$\hat{Y}_{t+1}^i = \sum_{j=1}^{|M|} \frac{\left[(1 - \chi_{j,(t+1)}^i) m_{j,t}^i \right]}{\sum_{j=1}^{|M|} (1 - \chi_{j,(t+1)}^i)} : \chi_{j,(t+1)}^i \in [0, 1], \forall j, t, i, \quad (17)$$

where $\chi_{j,(t+1)}^i$ is a normalized version of the recent loss of the model j on the random process i which computation is given by an evaluation metric of interest. A sample is considered *recent* if it arrived within the interval $[t - H, t]$. This mechanism allows a *blind* adaptation to drift by shifting the importance of the models in the final decision.

This methodology was exhaustively tested over data collected from three real-world case studies. Details about those datasets are provided in the following section.

4 DATASETS AND PREPROCESSING

The real-world data sources analyzed in this paper are taxi fleets where each vehicle broadcasts messages containing their status (*free/busy*) alongside their timestamped positioning using a GPS under the WGS84 format. These samples are broadcast with a regular frequency which is homogeneous within each fleet but may vary along the three Case Studies (CS). This section introduces each case study,

followed by the spatial and temporal discretization processes conducted to obtain a demand time series for each ROI within. Finally, descriptive statistics of each dataset are presented and discussed.

4.1 Case Studies

4.1.1 Porto, Portugal

Porto is a medium sized city with 1.3 M inhabitants located in Western Europe. Each vehicle out of a total of 438 within this fleet broadcasts, in average, one message per each 30 seconds. The analyzed dataset was collected between July 2013-July 2014, totaling roughly 1 Million trips. The city contains a total of 63 taxi stands to where the drivers are regulamentarily forced to drive after dropping a passenger off.

4.1.2 Stockholm, Sweden

Stockholm is the capital of Sweden. It is located in Northern Europe and it has 1 M inhabitants. The studied fleet contains roughly 1.6 k vehicles which broadcast messages with a frequency of 60 seconds. The studied dataset was collected between January 2014-January 2015, totaling up to 5.5 Million trips. The regulation schema is different from Portugal, not constraining the drivers to any particular decision after a drop-off.

4.1.3 Shanghai, China

Shanghai is one of the major cities in the world, inserted on an urban area with 34 M inhabitants. The dataset was originally processed in [23]. It comprises 12.3 M trips occurred between February 2006-March 2007 performed by a fleet of 5.4 k taxi vehicles. Similarly to CS B), there is no special regulation with respect to the directions to take after drop-off. On average, each vehicle broadcasts a message each 5 minutes.

4.2 Preprocessing

As preprocessing, the collected streams of Floating Car Data were discretized into time series using uniform time bins of P -minutes. A time series was generated for each ROI. In CS A), the ROIs were defined using the network topology given by the physical location of each stand. In the remaining CS, the space was discretized with a uniformly sized grid decomposition [6] with cells of 500×500 meters. Each pick-up event is considered to be dispatched from a ROI whenever it occurs in a radius of 100 meters. For simplicity, we considered that all three fleets operate in homogeneous 8h Daily Shifts (DS) (midnight to 8am, 8am-4pm and 4pm to midnight), even if this may vary a bit from their real operational schema.¹

The weather data was acquired through an open access repository [29]. We collected hourly summaries of temperature, precipitation and wind speed.

4.3 Data Analysis

Some statistics about the studied period in all cases are now presented. Fig. 2 shows the sample-based distributions of log-demanded taxi services per day type for case studies B

1. Note that the shifts are not used for any modeling component. The shift information is used for analysis purposes, namely, for results presentation and discussion in Section 5.

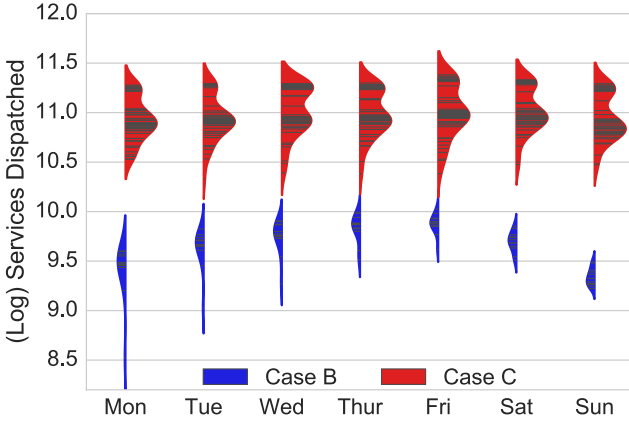


Fig. 2. Empirical demand distribution estimation.

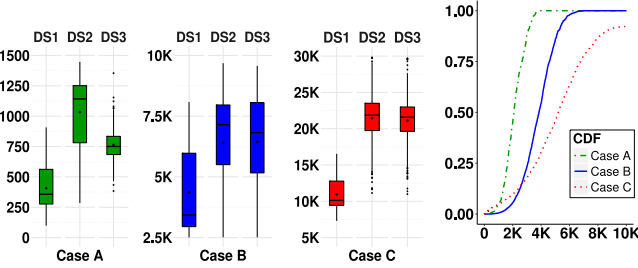


Fig. 3. Demand boxplots per day shift (DS) (one figure per CS) and empirical cumulative distribution function (CDF) of demand dispatched per driver.

and C—the ones with larger volumes of trips. Fig. 3 illustrates day shift boxplots of the demand, alongside the sample-based estimation of the Cumulative Distribution Function of the demand dispatched per driver.

Based on Fig. 2, we can conclude that there is a strong variation of the demand per daytype in Stockholm with unimodal structures—which contrasts clearly with the bimodal structure exhibited for CS C), thus illustrating a low variance within different weekdays. The boxplots in Fig. 3 confirm that such bimodal structure depicts the day/night seasonalities, where the sample-based mean of the dispatched services is 20 k services/shift—the double of the night ones. This fact highlights well the stochastic mobility behavior in CS C). The high variance of the demand within the night shifts in Stockholm is surprising, with large differences between mean, median and inter-quartile range. Porto concentrates its demand to working hours (DS 2), having a similar percentage of service dispatched per driver to Shanghai—while those are 30 percent lower than the ones exhibited in CS B). The mobility intelligence of each individual driver is more uniform in Porto compared to the two other CS—even if all of them exhibit a considerable variance. These facts highlight the potential real-world impact of BRIGHT on their operational profitability.

5 EXPERIMENTS

This section starts by introducing the test bed followed by our experiments, passing by the used evaluation metrics to then point out the obtained results as well as a brief discussion.

5.1 Experimental Setup

All the experiments were conducted using the R software [24] on a laptop with an eight-core 2.6 GHz CPU with 16 Gb

TABLE 1
Hyperparameter Setting

Hyperparameter	Description
α	Smoothing constant of FF-TVPP. Value found through a Tuned using a grid search over $[0, 1]$ with steps of 0.02.
ϵ_ω	Sets the memory limit of FF-TVPP. Fixed at 0.01.
T	Sliding window size to compute Arima/VAR models. Fixed at 2 weeks.
η	Learning rate of incremental training of the ARIMA/VAR parameters. Tuned using a grid search over $\{0.01a, \forall a \in \{1..20\}\}$.
k	Number of ROI to define a neighborhood in Drift-Aware VAR model. Tuned using a grid search over $\{2..9\}$.
ξ	Minimum similarity bound used to define a neighborhood in Drift-Aware VAR model. Tuned using a grid search over the values $\in [0.1, 0.9]$ with a step of 0.05.
ν_1	The confidence interval of the Hoeffding bound. Fixed to 95% [27].
ν, ϱ	Hyperparameters of the PH-test. Fixed accordingly to domain expert's suggestions.
H	Size of the Sliding-Window employed on the Two-Stage Ensemble. Tuned using a grid search over the values $\{1, 2, 4, 8, 10\}$.
β	Shrinkage factor on the L1-VARX model. Tuned using CTSCV. Test set size is kept stable but <i>rolling</i> in time while outdated test samples are appended to the training one.

of RAM. An aggregation period of $P = 30$ minutes was set (i.e., a new demand prediction is produced each 30 minutes). The weather data is also redundantly aggregated into a time series with the same periodicity of $P = 30$ minutes. The ARIMA, VAR and L1-VARX models orders were re-estimated each 24 h using a sliding window of two weeks.

Table 1 briefly describes the hyperparameter setting employed. All tuning procedures used the last month of the training set in each CS as validation set. The most sensitive hyperparameters ($\alpha, \eta, k, \xi, H, \beta$) underwent a tuning procedure before usage with a user-defined grid-search. The first 5 hyperparameters simply used the validation set to estimate generalization error of the models trained using all the possible combinations of the values within the grid. Then, using the best combination, β was tuned once using a validation set and a canonical time series cross-validation (CTSCV) procedure. Most of the remaining ones (i.e., $\epsilon_\omega, T, \nu_1$) were set following the recommendations of relevant literature on the topic [20], [26]. Finally, the hyperparameters related to PH-test (ν, ϱ) were calibrated with the help of domain expertise as they are sensitive to the concept of *anomaly*, where relevant change magnitude may vary across different urban scenarios. The authors recommend other users to follow similar procedures before applying this algorithm² to their own problem. The offline Model Selection procedure determined that the ARIMA/VAR processes were grouped together in one *family*, the L1-VARX in another family, while the two Poisson processes formed the third group for all CS. We selected only the most relevant ROI in terms of service volume to reduce

2. Source code publicly available at <http://tsbright.neclab.eu>.

TABLE 2
Experimental Results Using sMAPE

Method	Datasets				
	A	B	C	X	CS X (PeakHour PM)
FF-TVPP	17.28% ± 09.37%	21.46% ± 05.17%	22.26% ± 01.56%	15.73% ± 10.73%	21.03% ± 14.77%
ARIMA [18]	18.33% ± 11.04%	21.35% ± 06.76%	24.26% ± 02.85%	15.00% ± 11.44%	18.59% ± 13.32%
L1-VARX	18.36% ± 12.51%	18.93% ± 05.56%	22.09% ± 00.97%	11.26% ± 08.69%	15.00% ± 11.07%
Drift-Aware VAR	17.20% ± 10.18%	19.71% ± 06.08%	22.25% ± 01.06%	10.85% ± 07.86%	14.22% ± 10.35%
SoA Ensemble [20]	16.49% ± 09.38%	19.51% ± 05.60%	21.11% ± 01.52%	14.68% ± 11.09%	19.94% ± 15.32%
BRIGHT	16.09% ± 09.23%	18.56% ± 05.36%	20.89% ± 01.04%	11.88% ± 08.77%	15.93% ± 11.84%

The methods are grouped in three folds: univariate, multivariate and ensembles. BRIGHT systematically outperforms all the comparison partners from SoA and base learners. Datasets: A-Porto, B-Stockholm, C-Shanghai, and X-Synthetic with annotated drifts.

the computational effort of the test bed for CS B and C. The selection procedure was conducted over the validation set. The ROI containing less than the average of the sum of the demand quantities per ROI were discarded. As a result, CS B and C kept 178 and 210 ROIs, respectively.

5.2 Synthetic Data—CS X

This section describes a data generator specific for time series which is able to generate semi-synthetic datasets.

5.2.1 Synthetic Data

The generator is based on a simulator that recreates an industrial process of chemical production—the Tennessee Eastman Process (TEP) [25]. TEP is unstable, non-linear and controlled by Proportional Integral Derivative system for process stabilization [1]. This process is driven by 12 manipulable variables $\mathbf{u}_t$ and observed by 41 measurement variables $\mathbf{y}_t$ —thus creating 53 correlated processes. The simulator also includes 20 disturbances variables $\mathbf{d}_t$ (binary variables) to simulate process impairments and 12 desired set points $\mathbf{r}_t$ (desirable values to manipulable variables). The physical quantities represented by those are detailed in [2].

The drifts are produced by setpoints $\mathbf{r}_t$ variations. These variations may follow the patterns of either incremental, gradual or abrupt drifts. The time instants of the drifts are sampled from a Poisson process.

5.2.2 Semi-Synthetic Data

In order to generate datasets with similar characteristics to our original ones (i.e., semi-synthetic), we extended the above procedure as follows: first, we decompose both the original time series and the synthetic one

$$o(t) = o_s(t) + o_t(t) + o_r(t); s(t) = s_s(t) + s_t(t) + s_r(t). \quad (18)$$

$o(t), s(t)$ denote the original and the synthetic time series, while the s, t, r subscripts denote the decomposed seasonal, trend and random components of the signals, respectively. Both the perturbation of the original data as well as the induced drifts are made by recombining different components from both the original and synthetic data. The new semi-synthetic process $n(t)$ is obtained by modulation between the scaled components of the original process and the trend component of the synthetic process

$$n(t) = \left[a_0 s_t(t) \cdot [a_1 o_s(t) + a_2 o_t(t) + a_3 o_r(t)] \right], \quad (19)$$

where $|a| = 4$ denote a vector of scaling factors and $s_t(t) \in [0, 1], \forall t$. In this case, a was set as $a = 10, 1, 0.1, 2$. The intuition behind this choice is to smooth the original drifts and enhance the artificial ones, while keeping most of the natural dependence structure of the original signal. This process allowed to generate a new dataset (i.e., CS X), based on CS A.

5.3 Evaluation

The last 2 months of each dataset were considered as a test set for all CS. Naturally, the training/testing schema is *prequential*: when each sample's value is known, it is used to update our model's parameters during the test stage in an online learning fashion.

In terms of baselines, we compared two methods in the state-of-the-art (the ARIMA proposed in [18] and the State-of-the-art (S.o.A.) Ensemble from [20]) plus two offline baselines (FF-TVPP and L1-VARX) with our Drift-Aware VAR, and the overall framework, i.e., BRIGHT. To estimate each model's loss, we used a smoothed version of the Symmetric Mean Percentage Error by employing a laplacian constant. The final result is obtained by averaging all the singular metric values—as suggested by the related work [18], [20], [21]

$$sMAPE_t = \frac{1}{t \times K} \sum_{i \in \mathbb{K}} \sum_{j=1}^t \frac{|Y_j^i - \hat{Y}_j^i|}{Y_j^i + \hat{Y}_j^i + 1}. \quad (20)$$

5.4 Results

The results are presented over four distinct perspectives: 1) Table 2 encloses descriptive statistics of the experimental results on the prediction error per method/case study and during peak hours for CS X; 2) Fig. 4 depicts three illustrative examples on handling distinct drifts detection within the model selection framework; 3) Bias-variance analysis of the generalization error is illustrated in Fig. 5 for CS B and C, while a glimpse on the inductive variance's reduction obtained by BRIGHT is depicted in Fig. 7; 4) Fig. 8 illustrates an example of the updates on the neighborhood's topology of a given ROI throughout 24 h.

5.5 Scalability

All the base learners are *lightweight* models and their computation is completely independent from each other as well as among distinct ROI. Nevertheless, the experiments were conducted in a non-parallel fashion to conduct exact empirical evaluations of the algorithms' runtime. All base learners

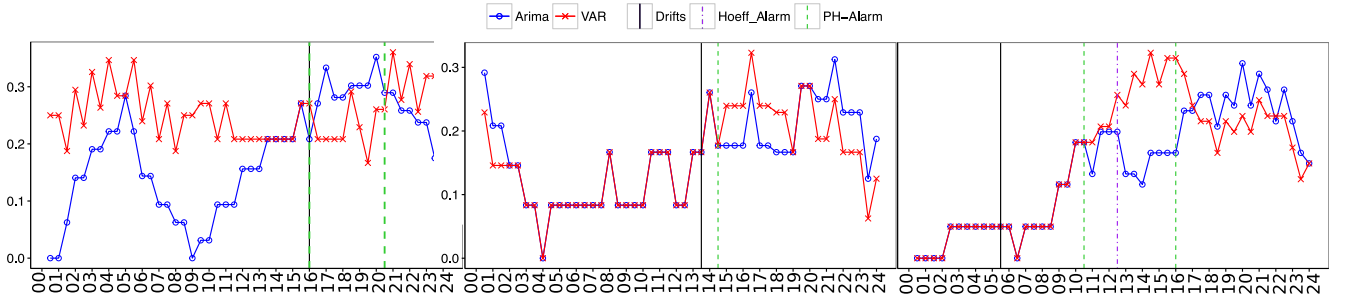


Fig. 4. Loss Illustration of two methods from the same *family* on different ROIs for 24h and its relation with different concept drifts, respectively, for CS X. X-axis denotes the time of the day (in hours) while y-axis denotes MSE. Note the PH-alarm in the first two figures dealing with Abrupt and Gradual drifts, respectively, and the Drift-Aware VAR model reaction to an incremental one in the latter.

took less than 1 s to either update their models or make new predictions for each ROI. The model order computations took, on average, 5 seconds and 3 seconds for ARIMA and VAR models, respectively. The Hoeffding bound test and the PH-test used less than 0.3 s to either update their sufficient statistics or to produce test results. A full prequential iteration cycle—i.e., updates and predictions for all the ROIs—took, on average, 11.2 minutes.

After initialization and hyperparameter tuning, BRIGHT’s spatial complexity depends on the memory of the base learners, since the ensemble model has a fixed-size memory. All memory sizes are fixed (e.g., the size of the validation set) or tuned from a finite grid. Consequently, the memory required to run this algorithm increases linearly with K —thus, logarithmically with the number of arrival samples, making it $\mathcal{O}(\log n)$ in worst case.

In terms of time complexity, many operators are incremental. The ARIMA/VAR regular model’s refitting may be costly as typical solvers include a Moore-Penrose pseudoinverse. Again, the number of elements in the matrix depends on the model’s order and on the neighborhood sizes. While the model’s order is assumed to be finite, the typical neighborhood sizes may increase logarithmically with K . Even if we claim that the algorithm’s worst-case time complexity is only $\mathcal{O}(\log n)$, we assume that it runs significantly better in practice.

5.6 Discussion

The discussion of the experimental results is made in four different folds across this section: (i) Generalization error and Performance, (ii) Drift Awareness, (iii) Bias-Variance Tradeoff, (iv) Significance test, (v) Potential Deployment and (vi) Limitations.

The (i) first conclusion drawn from Table 2 is that BRIGHT presents a lower generalization error than all other methods—even if the differences in CS A and B may look low in absolute terms. In fact, the main advantages of BRIGHT are uncovered during drift—and we cannot quantitatively compare the amount of drift of each CS. Yet, the results on CS X clearly exhibit a significant difference between the drift-aware methods—the base learner Drift-Aware VAR and the ensemble BRIGHT—when compared to its respective counterparts. In fact, we observed empirically that for periods with larger means (i.e., peak hours), the effects of drift on metrics reporting absolute error (e.g., SMAPE or RMSE) can grow even further. This observation is translated by the last column of Table 2, where the difference between BRIGHT and SoA is already on a ratio of 25 percent.

(ii) Figs. 8 and 4 illustrate well the informed adaptation to drift embedded in BRIGHT. In Fig. 8, the target stand denotes a University campus area, as well as the main city hospital. It hosts classes and medical emergencies during daytime, as well as regular academic nightlife events. In the north, the three stands highlighted in some figures correspond to boarding houses areas, while the stand highlighted in Fig. 8a and not in the remaining ones denotes student boarding houses as well. Finally, the stand highlighted only in Fig. 8d denotes an area covered by upper class private schools whose clients are likely to be immediate family (e.g., sons) of many of the employees of the institutions located in the target ROI.

Fig. 4 is key to our results: it shows examples of three distinct types of concept drift that occur in different ROIs. The detection of abrupt and gradual drifts is faster than the incremental one. Yet, the first PH alarm was triggered before the loss increase of the current *family winner*: VAR. In the same figure, a Hoeffding alarm is triggered later—flagging that the current neighborhood structure of this ROI changed

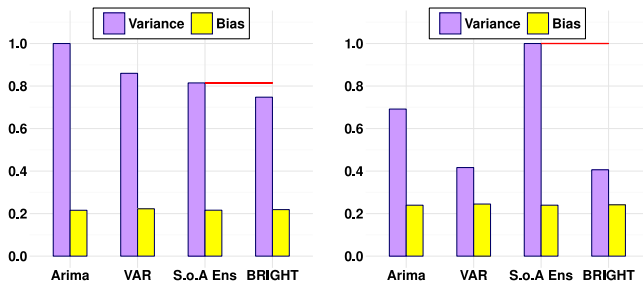


Fig. 5. Bias/Variance error for CS B (left) and C (right). X-axis denotes the different methods whereas y-axis denotes a normalized mean squared error (decomposed then on the bar plots).

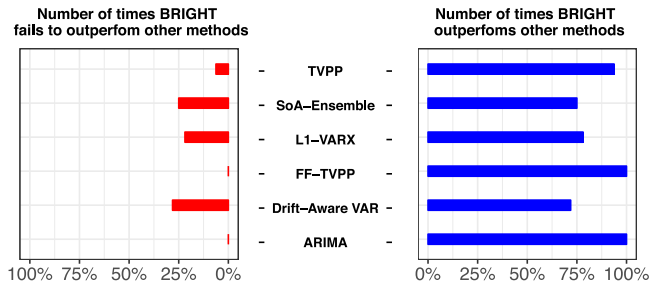


Fig. 6. Statistical significance test of the results using pairwise comparisons on a *BRIGHT-against-all*. Tests are conducted using the Wilcoxon signed-rank test over regular data intervals in the test set.

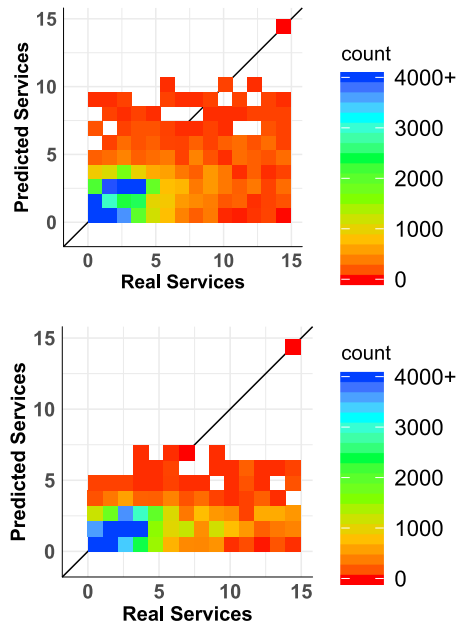


Fig. 7. Heatmap of predicted versus real services (y/x -axis) for SoA Ensemble/BRIGHT (top/bottom figs.) on CS X. Note the inductive variance reduction obtained by BRIGHT.

due to the occurrence of a drift. Consequently, its neighborhood is recomputed, the VAR model order/parameters are updated and the loss decays for its normal range. However, during the adaptation period, a fairly simpler model (ARIMA) was put in place (i.e., *winner*) to minimize the loss increase. This informed adaptation ability is a key point of BRIGHT.

Different components of this framework are responsible for handling different types of concept drift. Examples include the base learners and their optimization heuristics, i.e.: TVPP (incremental), FF-TVPP (gradual), ARIMA (abrupt), (L1-)VARX (both gradual and incremental) as well as the ensemble (both abrupt and reoccurring concepts). Also, the characteristics of drift may vary from occurrence to occurrence—even if the type is the same. Examples of the latter include the change rate or the concept component that is suffering the change (i.e., mean, co-variance or the entire functional form). Empirically, the illustrative results depicted in Figs. 8 and 4 show these abilities in practice.

(iii) Fig. 5 demonstrates empirically the significant reduction of the variance-type of error that BRIGHT achieves. The red horizontal line in Fig. 5 highlight the variance reduction when compared to SoA Ensemble—while maintaining a stable bias. This property is common to other popular ensemble schemas such as Bagging [11], [17]. Fig. 7 depicts how BRIGHT achieves such variance reduction. It does so by obtaining more short-tailed joint distributions which reduce the uncertainty around the expected value—turning more unlikely to sample (extreme) values from its tails.

(iv) The differences between the results depicted for the different methods in Table 2 seem reasonably small. This issue raises the question whether they are statistically significant. In a landmark paper, Demsar [7] proposed a unified frequentist methodology to evaluate different models over different datasets, including two statistical tests and a visual framework. A more recent publication of the same

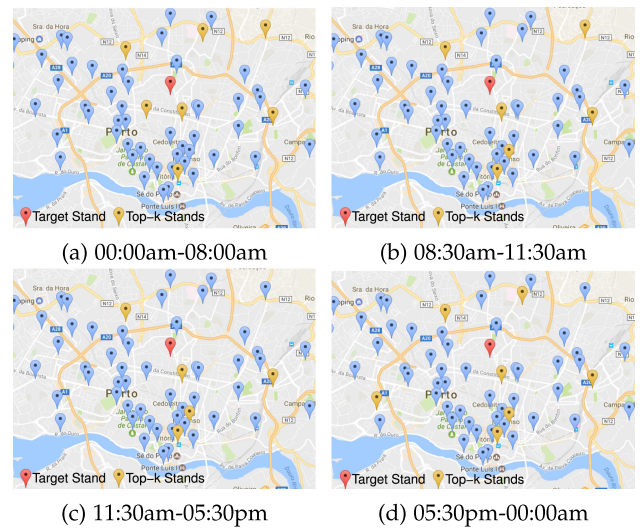


Fig. 8. Illustration of the dynamic neighborhood setting of the drift-aware VAR model in a ROI from CS A during 24 h.

author [3] proposes a Bayesian version of the methodology. However, all these tests assume stationarity of the underlying concept—and thus, they are not applicable to our testbed. Gama et al. [12] proposed an incremental test for the same purpose—the McNemar test, computed over a sliding window of time. Yet, McNemar test is computed over a confusion matrix—which is naturally only available for classification problems. Therefore, we have decided to employ the Wilcoxon signed-rank test over regular sequential data intervals of one week on the test set. Wilcoxon signed-rank test is a non-parametric test which provides a systematic way to assess if two samples are coming from the same distribution or not. The results depicted in Fig. 6 clearly highlights that BRIGHT systematically outperforms all its comparison partners (for a vast majority of the considered intervals).

(v) This prediction schema is able to produce on-demand predictions of service requests for short-term horizons independently of the bins definition due to the additive properties of histogram time series—as demonstrated in [21]. It can be deployed in a standalone fashion, as a *white box* decision support model where the drivers have explicit information about the future demand quantities to then decide which is the best strategy in a fully decentralized way—thus risking greedy *cannibalism* effects, well explained by the Nash equilibrium theory. Alternatively, its outputs can serve as input to any cost/utility function of a recommendation model which can be expressed probabilistically in the expectation to obtain near-optimal solutions rather than exact *global optima* (e.g., [31]).

Recently, a practical agent-based method to assess the potential impact of providing such predictive demand information to the drivers was proposed by Grau et al. [14]. In order to benchmark the potential impact of using BRIGHT versus the SoA Ensemble [20], we followed such methodology with CS A (Porto, Portugal). The obtained results pointed out that we can gain up to 4 percent on operational revenues in the evaluated period by disseminating the forecasted demand into 50 percent of the drivers. This gain comes mostly by the impact that more accurate predictions have on the passenger-finding strategies of more

conservative and inexperienced drivers—which end up reducing their vacant cruising time.

(vi) BRIGHT's simplicity lays the foundation for its strength, as well as for its limitations: its additive properties do not allow for capturing non-linear dependence structures; moreover, the short memory does not allow for learning with errors—namely, for reoccurring concepts to which the model's could be able to react faster if it had some sort of *drift history track* in memory—instead of informed by lagged alarms in real-time.

6 RELATED WORK

The related work in this topic can be folded into two categories: recommender systems [8], [13], [23], [28], [31], [32] and demand quantity forecasting [18], [20], [21], [30]. In recommender systems, the problem formulation is more complete whereas the researchers are not only able to model the historical service demand but also the drivers intelligence and their passenger-finding strategies. Typically, the idea passes by modeling the road network topology as a graph whereas the optimal route/direction with respect to profit is given by some sort of heuristic that aims to solve an optimization problem either approximately or in closed form. Early research focused on finding optimal directions [13]. Yuan et al. [31] went one step further by providing exact routes to of parking location on a subarea which would maximize the drivers short-term profit. Ding et al. [8] was one of the first works on this topic to raise the drift-related issues, which are handled with a blind adaptation schema.

The demand forecasting problem tries to mitigate the demand/supply imbalance issue by providing accurate predictions of the spatiotemporal distribution of the service demand. Li et al. [18] suggested a Seasonal ARIMA model to handle the problem computed on a re-training setting under a blind adaptation schema. Recent works in [20], [21], [30] suggest adaptive ensemble schemas similar to the one presented in Section 3.4.2 of univariate forecasting base learners. Hence, these schemas are limited to deal with just a few types of drift on a maximum admissible change rate [11].

7 CONCLUDING REMARKS

The maintenance of high quality, affordable and yet profitable door-to-door taxi service is a big challenge in major urban areas worldwide due to the mismatch often observed between service supply and a nearly stochastic demand. In this paper, we propose BRIGHT: a drift-aware supervised learning framework to predict taxi demand quantities in a given area for short-term horizons. BRIGHT is able to handle any type of real concept drift on the distribution describing the explanatory concepts of the citywide demand on an informed fashion. Moreover, it includes an Adaptive Ensemble schema that enforces diversity. On one of the largest experimental setups ever tried in this topic, BRIGHT achieved decreases in the generalization error up to 19 percent, with a variance-type of error reduction up to 50 percent. In the future, we plan to deploy this prediction mechanism in the real world to further validate it, as well as to improve its robustness.

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